# Haoming Li

haoming.li@utdallas.edu · +1 470 929 0691 · Homepage · Google Scholar · Github

## Research Interests

I work on **computer-use agents (CUAs)** for long-horizon tasks and **persona-aware CUAs**, in collaboration with MatrAIx. Currently, I'm interested in **persona AI** and **agentic memory**, which I expect persona agents to depend on. Earlier, I worked on scientific summarization and LLM planning.

## Education

**The University of Texas at Dallas**, Ph.D. in Computer Science, GPA 3.83/4.00 — Aug 2025 – Present
**Emory University**, M.S. in Computer Science, GPA 3.97/4.00 — Aug 2022 – May 2024
**Renmin University of China**, B.S. in Data Science and B.Econ. in Finance — Sep 2018 – Jun 2022

## Publications

[†]equal contribution

[1] **Haoming Li**[†], Defu Cao[†], Xu Hu[†], Yingtao Luo, Yiyang Wang, Jade Xu, Jiashu He, Wanting Jiang, Yifan Zhang, Yunhan Zhao, Siyang Wu, Chen Yang, Haixin Wang, Zheyuan Deng, Taoran Li, et al. *PersonaCUABench: Benchmarking Computer-Use Agents on Serving Diverse Personas.* Under review at **ACL ARR October 2026**.

[2] **Haoming Li**, Dingyi Kang, Xu Hu, Jinxin Yang, Zhu Xu, Yi Ge, Chenru Wang, Guanpeng Li, Miao Yin, Feng Chen, Bingzhe Li. *SAFE-Mem: Secure Agentic Memory with Fine-Grained Authority and Governance-Aware Retrieval.* Under review at **ACL ARR October 2026**.

[3] Xu Hu[†], **Haoming Li**[†], Yuxiang Wang, Jinxin Yang, Kangshuo Li, Licheng Wang, Dingyi Kang, Jiayu Song, Tianhao Wang, Zhanhua Hu, Xu Yang, Qiannan Li, Bingzhe Li, Feng Chen. *CUA-Recovery: Benchmarking Error Recovery in Long-Horizon Computer-Use Agents.* Under review at **ICLR 2027**.

[4] Siyang Wu, Weizhi Zhang, Chen Yang, Esther Xu Fei, Jorin Overwiening, Xu Pan, Jing Tang, Yixuan He, Kelvin C. Mo, Peiwen Li, Defu Cao, Xu Hu, Haixin Wang, Taoran Li, **Haoming Li**, et al. *PersonaScienceBench: Evaluating Scientific Agents with Diverse Simulated Researchers.* Under review at **ACL ARR October 2026**.

[5] Xu Hu, **Haoming Li**, Sabbir Ahmed, Md Nahiyan Uddin, Qiannan Li, Jessica Ouyang, Latifur Khan, Feng Chen, Bingzhe Li. *Toward Trustworthy Computer-Use Agents: Risk Propagation, Evaluation Gaps, and Human Governance.* Under review at **ACM Computing Surveys**.

[6] Weili Shi, **Haoming Li**, Sheng Li. *Enhancing Dynamic Text-Attributed Graph Modeling via Instruction Tuning.* **Findings of AACL-IJCNLP 2026** (to appear).

[7] **Haoming Li**, Jessica Ouyang. *GRASP: Graph-Reasoning Aided Survey Planning for High-Fidelity Related Work Generation.* **Findings of ACL 2026**. [paper]

[8] **Haoming Li**, Zhaoliang Chen, Songyuan Liu, Yiming Lu, Fei Liu. *Systematic Analysis of LLM Contributions to Planning: Solver, Verifier, Heuristic.* arXiv:2412.09666, 2024.

[9] **Haoming Li**, Zhaoliang Chen, Jonathan Zhang, Fei Liu. *LASP: Surveying the State-of-the-Art in Large Language Model-Assisted AI Planning.* arXiv:2409.01806, 2024.

[10] Yikun Wang, Rui Zheng, **Haoming Li**, Qi Zhang, Tao Gui, Fei Liu. *RESCUE: Ranking LLM Responses with Partial Ordering to Improve Response Generation.* **ACL 2024 Student Research Workshop**. [paper]

## Research Experience

**The University of Texas at Dallas**, Research Assistant (advisors: Feng Chen, Bingzhe Li) — Jun 2025 – Present

- **PersonaCUABench** (co-first author) [1]: selected and designed 44 personas across 11 scenarios, and evaluated state-of-the-art models on personalized CUA tasks for both objective task performance and user satisfaction.
- **SAFE-Mem** (first author) [2]: designed an agentic memory that accounts for both the stored content and its information source, making memory both secure and accurate.
- **CUA-Recovery** (co-first author) [3]: designed *ReRail*, the task and trajectory synthesis framework behind the benchmark (2,250 human-verified erroneous states for testing, 8.6K trajectories for training).
- Analyzed six agents at six takeover depths (best error-awareness rate: 44.0%) and fine-tuned Qwen3.5-35B-A3B on the training split with SFT, raising its error-awareness rate from 19.9% to 35.2% on held-out states.
- **Trustworthy CUA survey** (second author) [5]: reviewed and organized the literature on CUA security, privacy, reliability, controllability, and accountability.
- **GRASP** (first author, advised by Jessica Ouyang) [7]: designed, implemented, and evaluated a graph-reasoning planner for related-work generation.

**University of Virginia, Reasoning & Knowledge Discovery Lab**, Research Assistant (advisor: Sheng Li) — Jun 2024 – May 2025

- Instruction-tuned LLMs to encode and reason over dynamic text-attributed graphs [6].

**Emory University, NLP Group**, Research Assistant (advisor: Fei Liu) Mar 2023 – May 2025

- **LLMs for planning** (first author) [8, 9]: led an empirical study of LLMs as solver, verifier, and heuristic function in planning, and a survey of LLM-assisted planning.
- **RESCUE** [10]: trained a response ranker from partial orderings over candidate LLM outputs to improve response generation.

## Service and Skills

**Reviewer:** NeurIPS 2026, IEEE BIBM 2026. **Languages:** English (fluent), Chinese (native).
**Programming:** Python, PyTorch, C/C++, SQL, Bash. **Tools:** Hugging Face Transformers, vLLM, Docker, Spark.